

Pulsed Nuclear Resonance Apparatus

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Pulsed Nuclear Resonance Apparatus*

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The description of a complete pulsed nuclear resonance spectrometer suitable for experiments in solids is given. Its major features are the ability to operate at any frequency between 2 and 30 Mc, an rf pulse output power of 5 kW over most of this range, a receiving system recovery time of 5–10 μ sec at 10 Mc, a sensitivity usually limited by thermal noise in the receiver coil, use of phase coherent detection, the availability of a wide variety of pulse sequences, and good long and short term stability of the entire system. Brief instructions for operating the apparatus are included. A method of using the "boxcar" integrator to record directly the steady-state absorption (χ'') or dispersion (χ') is explained.

I. INTRODUCTION

IT is the purpose of this article to describe the construction, performance, and operating procedure of a complete pulsed nuclear resonance apparatus.

This apparatus is a further development of one constructed several years ago,¹ and has been in operation for about two years. It has been designed for nuclear magnetic and pure quadrupole resonance work in solids, where the typically broad lines require short, intense bursts of rf power and fast receiver recovery times. Major features of the apparatus are (a) the ability to operate at any frequency between 2 and 30 Mc, (b) an rf pulse output power of 5 kW over most of this range (leading to pulsed circularly polarized rf magnetic fields on the order of 50 G), (c) a receiving system recovery time of 5–10 μ sec when operating at 10 Mc, (d) a sensitivity usually limited by thermal noise in the receiver coil, (e) the use of phase coherent rf detection, (f) the ability to generate a large variety of pulse sequences, and (g) good long term stability of the entire system.

In what follows, it will be assumed that the reader is familiar with the major features of pulsed nuclear resonance.²

Section II provides a detailed description of the apparatus and Sec. III describes its use. Included in Sec. III is a brief description of a method to record directly the steady-state resonance absorption $\chi''(\Omega)$ and dispersion $\chi'(\Omega)$ using a "boxcar" integrator. This development makes it convenient to use the apparatus to search for magnetic resonances.

II. DESCRIPTION OF THE APPARATUS

A. Introduction

An ideal pulsed nuclear resonance apparatus would tip the nuclear spin system with a huge rf pulse that lasts for a

very short time interval, and then instantly be ready to record a short-lived free induction decay signal without adding any noise or distortion to it. In practice, one approaches this ideal by producing as large an rf field as possible, turning it on and off as quickly as possible, and having a low noise linear receiver recover from the overload of the rf pulse as soon as possible. These general considerations plus the desire of adaptability to many classes of experiments led to the design described here.

This section is devoted mainly to describing individual parts of the apparatus. Most of these descriptions will include a statement of the more important requirements of the individual part, a description of special steps taken to meet these requirements, an outline of the operational principles of the part, and a statement of its level of performance.

B. Block Diagram

The block diagram in Fig. 1 shows the relationship of the various parts of the apparatus and includes some of the more important waveforms.

A Tektronix 190B constant amplitude sine wave generator is the rf source (master oscillator). Its well-regulated output voltage is split into two channels; one passes through the phase shifter, rf gate, and power amplifier, while the other goes via an attenuator to an input of the receiver chassis. The phase shifter permits control of the phase difference between the transmitter rf pulse and the reference signal which is fed to the receiver.

Groups of pulses to operate the rf gate originate in the pulse generator. Accurate spacing between pulses in a given group is governed by timing markers locked to the period of the square-wave generator. Radio frequency pulses from the rf gate are amplified by the transmitter, which drives the transmitter coil in the crossed coil assembly. (Other coil arrangements can be used, as explained in Sec. II K.)

The free induction decay signal and rf pulse overload are

* This work was supported by the National Science Foundation.
¹ W. Gilbert Clark, thesis, Cornell University, 1961 (unpublished); U. S. Office of Naval Research, Technical Report No. 5, Contract N-ONR-401(15), June 1961 (unpublished).

² For the reader who is unfamiliar with this kind of experiment, the following references should form an adequate introduction: E. L.

Hahn, *Phys. Rev.* **80**, 580 (1950); E. L. Hahn, *Phys. Today* **6**, 4 (1953); H. Y. Carr and E. M. Purcell, *Phys. Rev.* **94**, 630 (1954); and I. Solomon, *Phys. Rev.* **110**, 61 (1958).

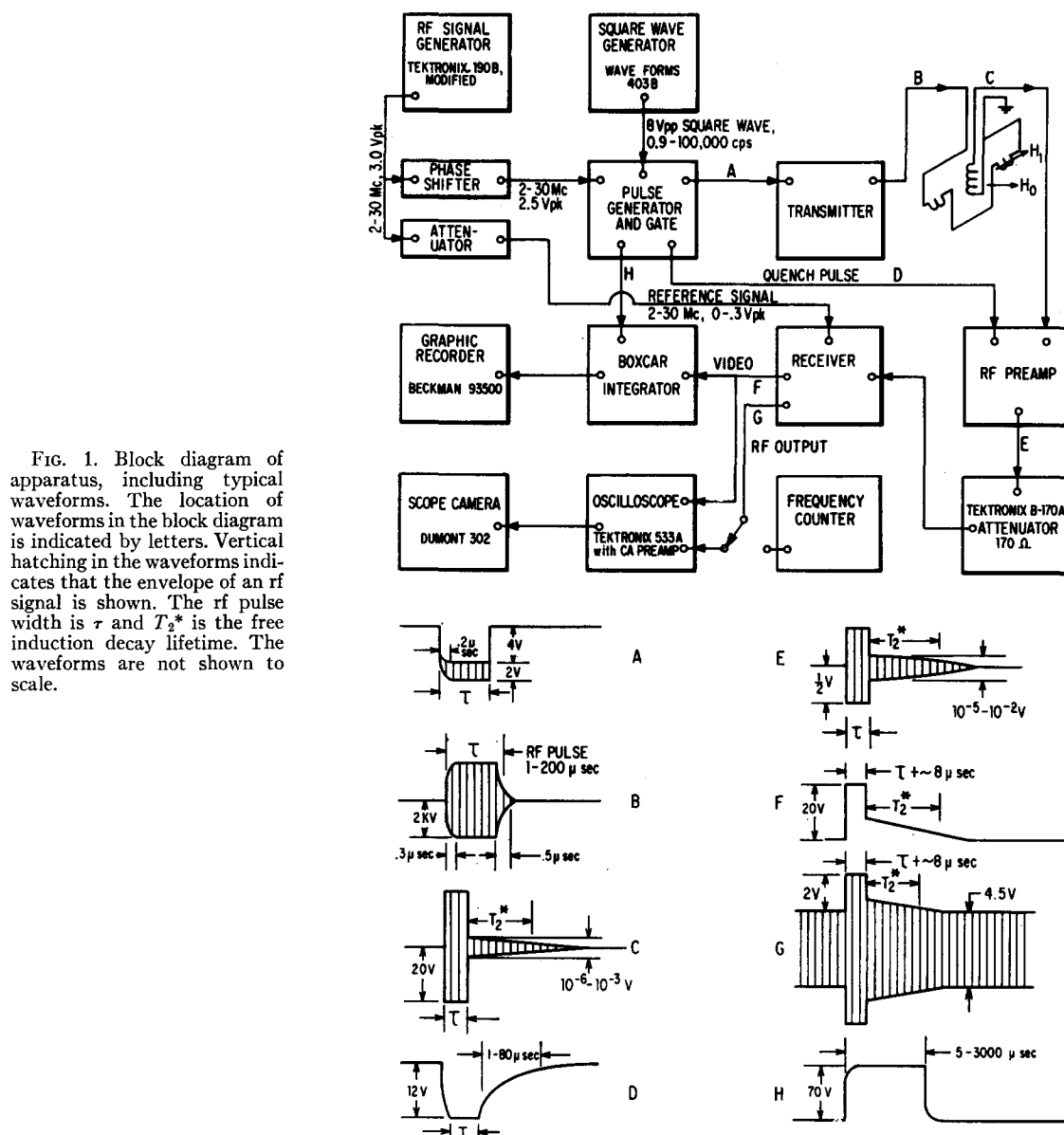


FIG. 1. Block diagram of apparatus, including typical waveforms. The location of waveforms in the block diagram is indicated by letters. Vertical hatching in the waveforms indicates that the envelope of an rf signal is shown. The rf pulse width is τ and T_2^* is the free induction decay lifetime. The waveforms are not shown to scale.

picked up by the rf preamplifier from the receiver coil. A special pulse from the pulse generator (quench pulse) helps to speed the recovery of the receiver coil from the overload of the rf pulse. The quench pulse does this by lowering the Q of the receiver coil until the ringing induced by the rf pulse has stopped, then raising the Q to record the nuclear resonance signal.

An attenuator between the rf preamplifier and the receiver allows accurate changes in over-all rf gain to be made up to a factor of 1000:1.

In the first stage of the receiver, the free induction decay signal and the reference signal are added to make possible phase coherent detection. Since the reference signal has a steady amplitude (which is adjusted to be large compared to the nuclear resonance signal), its rectified amplitude can

be used to provide automatic gain control (AGC) of the receiver. In this mode, the magnitude of the receiver gain is controlled by the attenuator in the reference signal channel.

The receiver has two outputs; one is the undetected nuclear resonance-reference signal, and the other is the detected (video) envelope. These are displayed (often simultaneously) on an oscilloscope screen, from which photographs can be made. The rf output of the receiver is also suitable for driving a frequency counter.

A boxcar integrator can be used to sample the video output of the receiver. It is gated by a pulse from the pulse generator, and its output drives a graphic recorder.

The entire apparatus is run from a line voltage regulated to 0.1%.

C. Master Oscillator

The master oscillator is required to cover a range of 2–30 Mc with good amplitude and frequency stability. Since this signal appears at the detector when phase-coherent detection is used, any spurious amplitude or frequency modulation will appear as an unwanted signal or noise. A Tektronix 190B constant amplitude sine wave generator, modified as in Fig. 2, meets these requirements. The modification reduces incidental amplitude modulation to 4×10^{-5} ; at the receiver output this signal is small compared to noise for all conditions of receiver gain and bandwidth. Noise output and amplitude drift of the master oscillator are negligible. Incidental frequency modulation is less than 1 part in 10^5 , and the diurnal frequency drift in our laboratory is about 1 part in 10^4 . After an initial warmup period, the frequency drift is usually less than 1 part in 10^5 over a period of 1 h.

Several steps have been taken to reduce rf leakage from the signal generator. (These are not shown in Fig. 2.) A shielded rf filter has been added to the 115-V ac line cord at the point where it enters the chassis. The side panels have been attached more firmly with self-tapping screws. A grounded shield has been added to the cable connecting the output attenuator and the main chassis. Finally, bypass capacitors to ground have been added as follows: $0.03 \mu\text{F}$ to each of the meter leads, $0.1 \mu\text{F}$ across the 6C4 filament, and $0.02 \mu\text{F}$ each to pins 1, 2, and 3 of the cable connector at the main chassis.

Since operation of the 190B stepped-attenuator causes slight frequency changes, this attenuator is never varied from its maximum output position. The continuous amplitude control is set for an output of 6.0 V peak-to-peak, which is sufficient to operate the phase shifter-attenuator circuit.

D. Phase Shifter-Attenuator

The main requirement of the phase shifter is that it provide at least 180° of phase shift between 2 and 30 Mc. This is easily accomplished with continuously variable rf delay lines. The amplitude stability required of the refer-

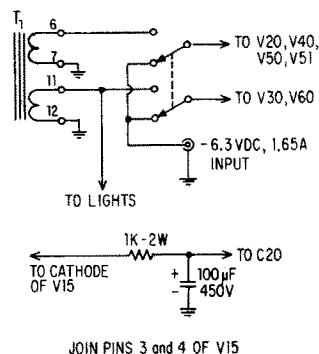


FIG. 2. Modifications to the Tektronix 190B master oscillator.

ence channel is obtained by using only a cathode follower and passive components.

The circuit diagram for the phase shifter and attenuator appears in Fig. 3. The $120\text{-}\Omega$ input load and cathode follower buffer amplifiers prevent control variations from changing the frequency of the Tektronix 190B master oscillator. In the reference amplitude control channel there is sufficient variation in attenuation to provide reference amplitude controlled gain over a range of about 150 to 1. Terminating the output with a matched $50\text{-}\Omega$ cable makes the attenuator characteristics relatively independent of frequency.

The phase shift channel uses two delay lines for control of the phase difference between the rf pulses and the reference signal. Terminating resistors have been chosen empirically to minimize the VSWR in this circuit over the frequencies of interest. For all frequencies at which it has been checked, the VSWR is less than 1.3. Since the output of the transmitter is independent of input voltage variations in this range, no further attempt has been made to reduce this VSWR.

E. Pulse Generator

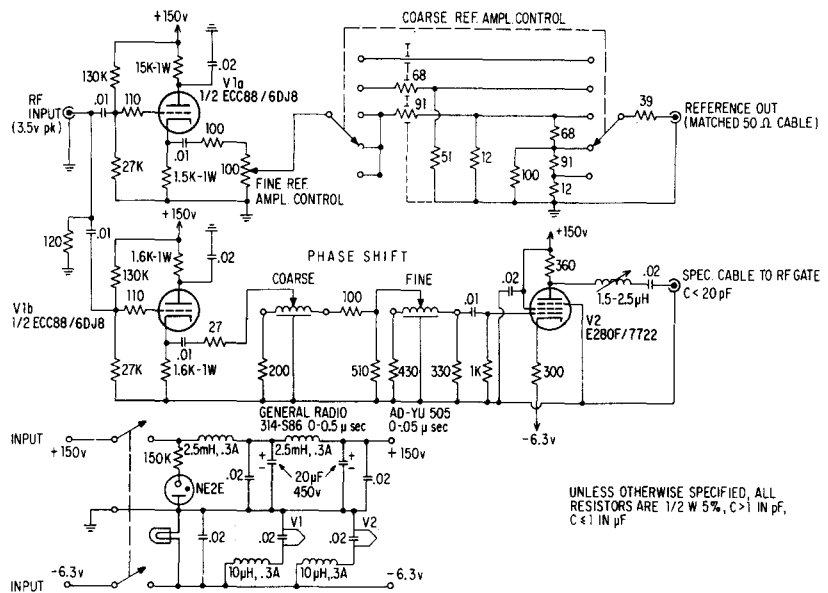
The pulse generator is required to produce groups of pulses in a wide variety of configurations with precise and stable control of their widths and their separation within a given group. It has been possible to satisfy these requirements by controlling the pulse widths and separation between groups with conventional multivibrator circuits. Control of the major portion of the interval between pulses of a group is performed with an audio oscillator and simple counter circuit.

A circuit diagram for the pulse generator, which includes the rf gate, is shown in Fig. 4. The square-wave output from a Waveforms 403B generator is differentiated, inverted, and fed to a diode gate at V1. If this gate is open, each positive transition of the square-wave input causes the bistable multivibrator V2 to switch. This then advances the current in the beam switching tube V3³ one position in its cycle of ten steps. Each target of V3 is connected through an RC differentiator to one contact of switches S2, S6, S9, and S12. A negative pulse passing any of these switches triggers the corresponding pulse channel. Thus, in any of the four pulse channels, the arrival of current at a given target of V3 can generate a trigger signal that is in synchronism with the square-wave generator.

Channel 4 inserts a delay between successive 10-step cycles of V3 by controlling the gate at V1. If the monostable multivibrator (V4 and V5) is in its stable position, the gate at V1 is open and V3 is allowed to switch. When the target current of V3 reaches the position chosen by

³ The BX 3000 is manufactured by the Burroughs Corporation, Electronic Components Division, Plainfield, New Jersey.

Fig. 3. Phase shifter-attenuator circuit.



S12, V4-5 switches to the astable position for a time interval determined by the positions of S14 and the 10-k Ω potentiometer in the grid of V5. During this interval, the gate at V1 is closed and V3 does not advance. This action causes the ten-step cycles of V3 to be separated by the duration of V4-5. To insure that a transition of V4-5 does not overlap a positive transition of the square-wave input, the latter is differentiated and fed to a grid of V5 as a synchronizing signal. This signal forces the duration of V4-5 to end upon a negative transition of the square wave.

If no pause between 10-step cycles is wanted, S12 is advanced to position NC (no connection), which disconnects the trigger signal to V4-5. If, on the other hand, arbitrarily large delays between 10-step cycles of V3 are needed, S12 is set to the manual position and the 10-k Ω potentiometer tap at V5 is adjusted to give a maximum voltage. In this configuration, V4-5 is bistable, and each manual closure of S13 triggers one 10-step cycle in V3 (provided the frequency driving V3 is not too high—see end of this section).

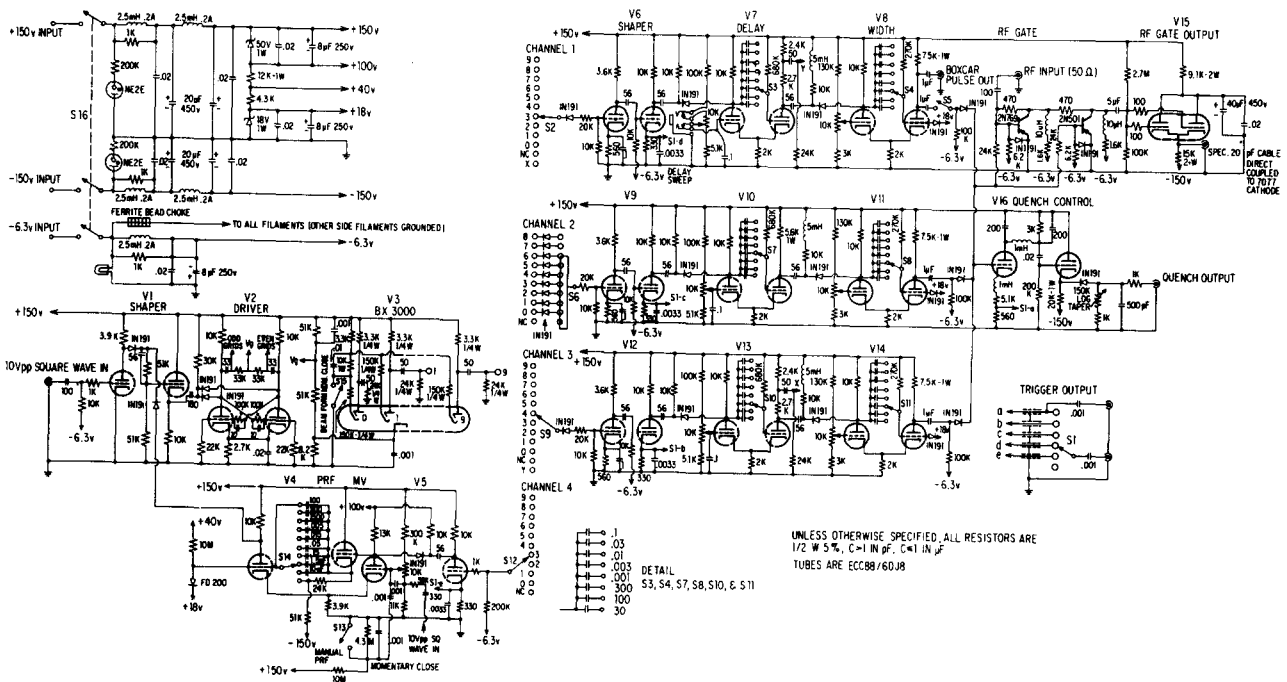


Fig. 4. Pulse generator circuit.

When the pulse generator power is first turned on, no target current flows in V3. To form a beam current, it is necessary to lower the potential of one spade temporarily. This action is accomplished by closing and opening S15. After that, V3 is ready to operate.

Pulse channels 1, 2, and 3 are used to operate the rf gate. Channel 1 also has an external output (for gating the boxcar integrator) and the option of being disconnected from the rf gate by S5. Switch S2 selects the trigger source for channel 1, which can be any target of V3 or the trailing edge of the delay pulse from V13 in channel 3 (position "X"). The latter provision is useful for certain applications of the boxcar integrator. If S2 is placed in position "NC," no pulse is generated in Channel 1.

The output of Channel 1 is a positive pulse whose width is determined by the time constant of monostable multivibrator V8. Stepwise variation of this width is governed by S4, and continuous variation is obtained with the 10-k Ω grid bias potentiometer. The timing of the output pulse can be varied independently by the delay multivibrator V7. A provision for external sweeping of this delay has been included for certain boxcar integrator applications. For this mode of operation, the 10-k Ω potentiometer is simply replaced with an externally driven one, the operation proceeding via a phone plug. Tube V7 also provides a trigger output Y for Channel 3.

Channel 2 has the same delay and width controls as Channel 1, but has no external output and, when triggered, always operates the rf gate. The unique feature of Channel 2 is that S6 allows it to be triggered by one to nine consecutive positions of V3, or not at all. Channel 2 is most useful for generating up to nine equally spaced rf pulses of identical width.

Channel 3 differs from Channel 1 only in the omission of an external delay sweep option, an external width output, and an rf gate disconnect switch analogous to S5.

In the absence of an rf gate pulse, the gate transistors are biased to give a collector current large enough so that their collector-to-emitter impedance is very low. When in this state, they present a small rf impedance to ground, and the rf signal from the rf input is attenuated in the 470- Ω resistors. During the time that any of the pulse channels directs a positive pulse to the transistor bases, they are biased to cutoff, their collector-to-emitter impedance is large, and the rf signal is passed to the rf gate output cathode follower (V15). The transistor cutoff also sends a -6-V pedestal to V15. The output from V15 is then a pulse of rf superimposed on a negative pedestal. This signal, when direct coupled to V1 of the transmitter, produces the result described in Sec. II-F. Base-to-emitter clamp diodes have been included to prevent damage to the transistors.

The same pulse that drives the rf gate transistors also drives the quench control (V16). An inversion stage pro-

duces at the cathode follower section a negative pulse which lasts for the duration of the gate pulse. At the end of the rf pulse, a diode disconnects the quench output from the positive recovery of the cathode follower. The recovery time at the quench output is then governed by the 150-k Ω logarithmic taper rheostat and the 500-pF capacitor. For the duration of the rf pulse, this waveform lowers the Q of the receiver coil. When the quench output waveform reaches a potential of about -2 V, the receiver coil Q is raised to its normal value. Some rf filtering has been included in the quench control circuit to prevent the rf voltage from bypassing the gate.

A trigger output switch has been provided to make available the important timing pulses for triggering an oscilloscope.

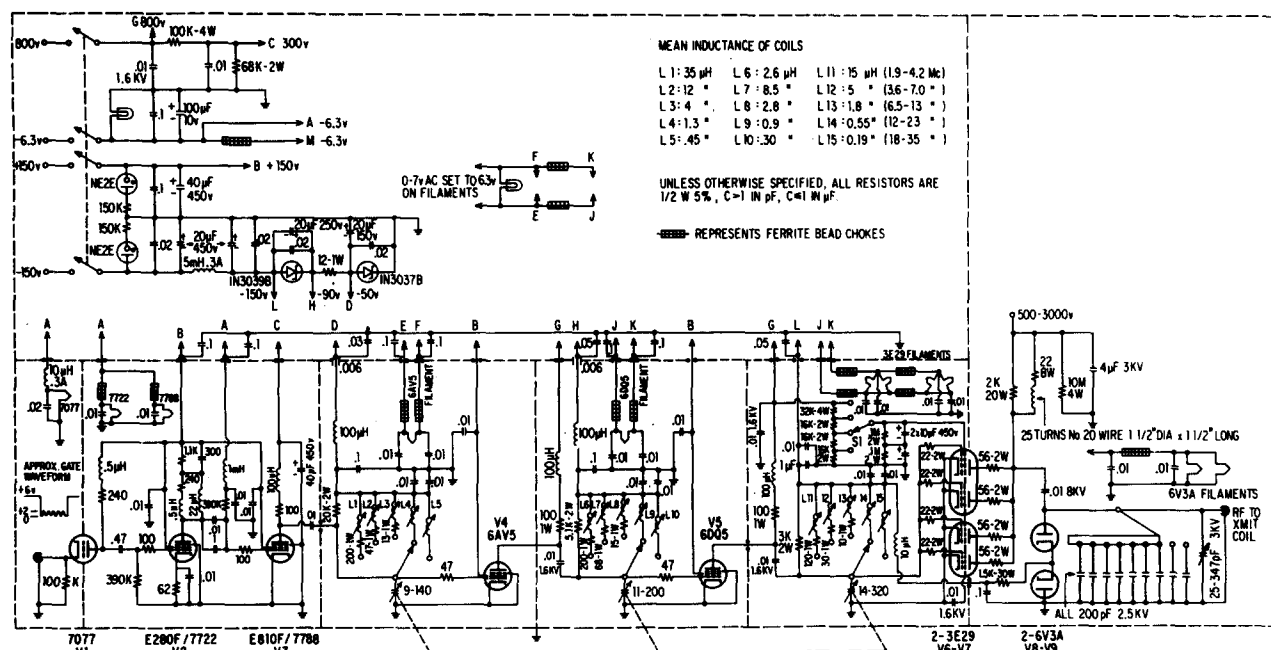
Channels 1, 2, and 3 have identical performance. The duration of the delay multivibrators can be varied between 3 μ sec and 30 msec, and the duration of the width multivibrators between 1 μ sec and 10 msec. At the outputs of these circuits, the rise and fall times are about 0.15 μ sec. When the external boxcar sweep option in channel 1 is used, the linearity of the delay with respect to external potentiometer settings is 1.5% over a delay range of 4:1. If channel 4 is not operated in the manual mode, its delay is adjustable from 3 msec to 40 sec.

The carrier suppression at the output of the rf gate has been measured at 10 and 20 Mc. In either case, it exceeded a factor of 200.

Short term variations in the duration of the various timing circuits (jitter) has not been measured. It has, however, been clear during the performance of many experiments that the jitter in these circuits is not more than 1%, and is probably much less. The same remarks apply to long term variations (drift) if the limit is set at 5%.

When short rf pulses are used, the jitter in rf pulse width at the output of the transmitter will exceed the jitter in the waveforms driving the rf gate because the leading edge of the gate driving waveform is not locked to the phase of the master oscillator signal. This lack of registry between these waveforms leads to a variation of not more than one cycle of rf in the width of the transmitter output. Thus, a pulse containing 20 cycles of rf will exhibit a jitter in width of roughly 5%.

The maximum frequency with which V3 and its driver circuitry will operate is about 10^5 cps. If, however, channel 4 has been set for one of its larger delay ranges, it will not operate quickly enough, and V3 will go through several 10-step cycles before stopping. Therefore, when operating with long intervals from channel 4, it may be necessary to restrict the square-wave generator frequency to less than 10^4 cps. The same precaution applies when channel 4 is operated in the manual mode. Ordinarily, the occurrence of this multiple cycling will be obvious when viewed on the oscilloscope.



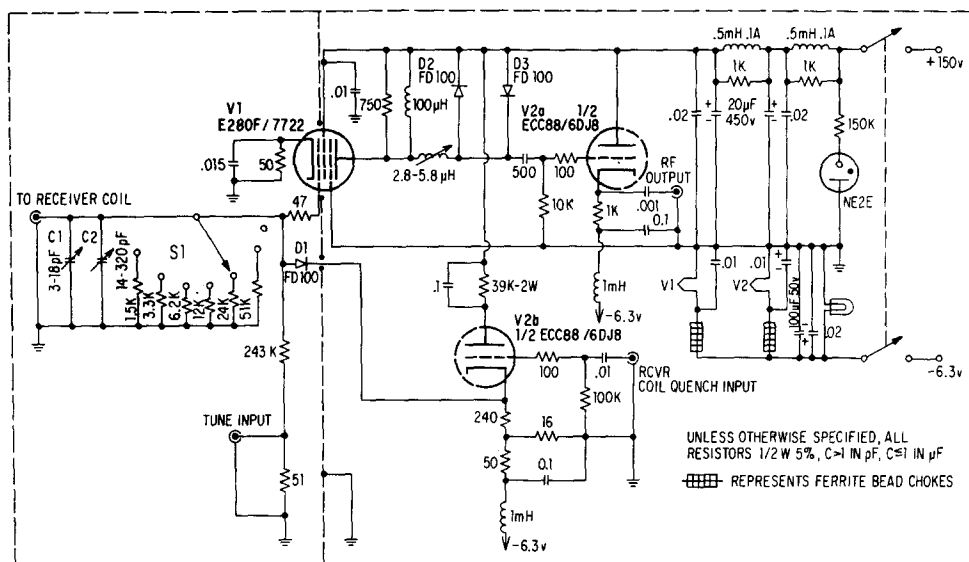


FIG. 6. Preamplifier circuit.

in this way. Additional reduction can be achieved with an external load resistor. A 51- Ω 2-W resistor has been used frequently for this purpose.

Several precautions should be observed with the transmitter. The pulse duty factor should never be run high enough to make the average power dissipation in V6 and V7 exceed 30 W. Furthermore, these tubes should be tested to insure that in this circuit their quiescent plate current does not exceed 0.1 mA. If it does, their noise output may couple strongly enough to the receiver to become a nuisance. Finally, ceramic capacitors should not be used for coupling or bypass applications in the last rf stage. Their piezoelectric dielectrics may cause a spurious ringing signal which can persist long after the rf pulse has ceased.

The maximum power output of the transmitter during rf pulses is approximately 5 kW from 2–20 Mc, and drops to 1.8 kW at 30 Mc. Below 10 Mc, the transmitter output impedance with 3.0-kV plate input voltage is about 250 Ω . The transmitter will handle pulses up to at least 200 μ sec in length with less than 10% sag. (Ordinarily, pulses from 1–30 μ sec in length are used.) Pulse rise and fall times depend somewhat on the receiver coil loading, and are typically on the order of 0.3 μ sec.

An important specification of a pulse transmitter intended for nuclear resonance work is the ratio of the amplitude of an rf pulse to its steady rf output in the absence of a pulse (rejection ratio β). The rf feedthrough will be negligible, i.e., perturb measurements by less than 1%, as long as the saturation factor $S = \gamma^2 H_1^2 T_1 T_2^*$ is less than 0.01.⁵ T_2^* is the free induction decay lifetime, T_1 the nuclear spin-lattice relaxation time, H_1 the amplitude of the circularly polarized rf field, and γ the nuclear gyro-

magnetic ratio. If, for example, we assume a 90° pulse length of 0.1 T_2^* , the restriction on T_1 becomes $T_1 < 4 \times 10^{-5} \beta^2 T_2^*$. A lower limit on β has been measured at 10 Mc by connecting the receiver directly to the transmitter in the absence of rf pulses and operating with conditions that would give 2 kV for the amplitude of rf pulses. The absence of any observable rf feedthrough placed an upper limit of 1 μ V on leakage to the transmitter coil; hence, $\beta > 2 \times 10^9$. Even if T_2^* is as short as 10 μ sec, our conditions require only that T_1 be shorter than 1.6×10^9 sec to avoid saturation by rf feedthrough. This restriction is negligible.

G. Preamplifier

The preamplifier (Fig. 6) is required to cover a wide frequency range, have high sensitivity, and recover quickly from large overload signals. A wide-band circuit meets the first requirement. Use of a high g_m tube at the input provides low input noise. Fast overload recovery is insured by direct coupling the receiver coil to the input stage and crossed diode limiting at the plate, which prevents overloading the stage that follows.

V1⁶ supplies the gain of the preamplifier. Even though the tube is a pentode, its measured equivalent series shot noise resistance at 10 Mc in this circuit is only $220 \pm 20 \Omega$. This parameter should represent the noise characteristics of the preamplifier from 2–30 Mc quite well. Since the source impedance of the receiver coil circuit is ordinarily many times this value, the ultimate sensitivity of the apparatus is usually determined by thermal noise in the receiver coil. An exception can occur if the temperature of the receiver coil is low enough (see Sec. II K). The plate

⁵ For a definition of S and an explanation of its significance, see N. Bloembergen, E. M. Purcell, and R. V. Pound, Phys. Rev. 73, 679 (1948).

⁶ The 7722/E280F is available in the United States through Siemens America Incorporated, 350 Fifth Avenue, New York 1, New York.

load of V1 has series peaking to extend its bandwidth and crossed diodes (D-2,3) to clip signals above 0.5 V of either polarity and prevent overloading of subsequent stages. When the cathode follower V2a is loaded with 170 Ω , the over-all small signal rf gain of the preamplifier is about 10 from 2–35 Mc.

Q switching of the receiver coil, the reason for which is explained in Sec. II K, is accomplished by V2b and D1. With no signal input to V2b, D1 is reverse biased and exerts no loading on the receiver coil. But when a signal brings V2b to cutoff (waveform D of Fig. 1), D1 is forward biased by 1.5 V through 240 Ω , and the Q of the receiver coil is severely reduced. The duration of the trailing edge of waveform D (provided by the pulse generator) is adjusted to lower the Q of the receiver coil only until the ringing from rf pickup out of the transmitter coil has ceased. Then, V2b resumes conduction, and the receiver coil Q is raised for recording the free induction decay signal.

Coil loading resistors controlled by switch S1 have been added for measuring signals with characteristic times too short to be recorded by an undamped receiver coil.

To facilitate tuning the receiver coil, the tune input jack has been provided. For this function, the reference signal is removed from the receiver and connected to the tune input. The rf output of the receiver is monitored and the gain of the receiver set below the AGC threshold. When

C1 and C2 are adjusted to give a maximum rf output within their tuning range the receiver coil is tuned.

H. Receiver

The receiver must produce stable gain and low distortion over a frequency range of 2–30 Mc. Tuned amplifier stages with switched coils and variable capacitors are used to cover the frequency range. Low distortion is achieved through phase sensitive detection, which also permits use of the boxcar integrator for optimum noise reduction⁷ and recording $\chi'(\Omega)$ and $\chi''(\Omega)$. Excellent gain stability is accomplished by using the reference signal for AGC.

The first three stages of the receiver (Fig. 7) are capacitor tuned rf amplifiers with band-switched inductors. The coils plus trimmer capacitors C1, C3, and C5 are adjusted for optimum tracking. Since resistive loading in the plate circuits sets their bandwidth at about 0.5–2 Mc, tracking adjustments are not critical. Crossed diodes in the plate loads of the first two stages limit their outputs to about 0.5 V; hence, serious overloads never occur in these stages, even though they may be present at the input of the preamplifier.

The phase coherent detection reference signal is applied to the cathode of the first rf stage. After detection, the rectified reference signal is applied as AGC bias to the first two stages. With AGC control, the peak rf at the detector

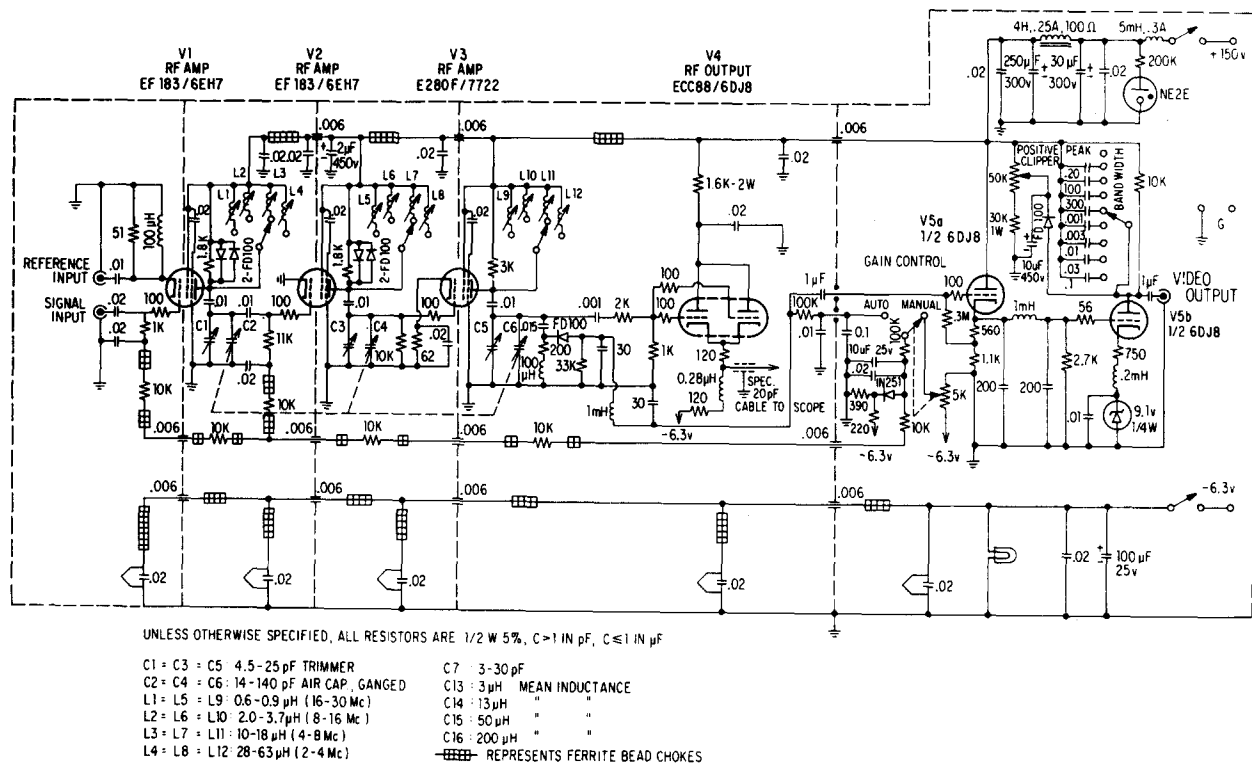


FIG. 7. Receiver circuit.

⁷ D. F. Holcomb and R. E. Norberg, Phys. Rev. **98**, 1074 (1955).

The maximum rf gain of the receiver is about 2×10^3 , and the gain of the video stage is 7. With the range of reference signal amplitudes available, it is possible to vary the rf gain continuously over a factor of 150 to 1. Further gain variations in small increments up to a factor of 10^3 to 1 with an accuracy of about 1% are provided by the attenuator between the preamplifier and the receiver. With the receiver gain set to the manual mode, the short term drift in gain of the preamp and receiver has been measured as high as 2%/h. The gain stability of this combination with the receiver in the AGC mode is much better. A care-

The bandwidth of the receiver is determined by (a) the bandwidth of the rf stages, (b) the bandwidth of the detector, and (c) the bandwidth of the video stage. Variation in tuning capacitance adjusts the bandwidth of the rf stages between 0.5 Mc at the low end of a band to 2 Mc at the high end. These are the equivalent of 0.25 and 1 Mc in upper cutoff frequency when converted to zero frequency (i.e., detected). The upper cutoff frequency of the detector

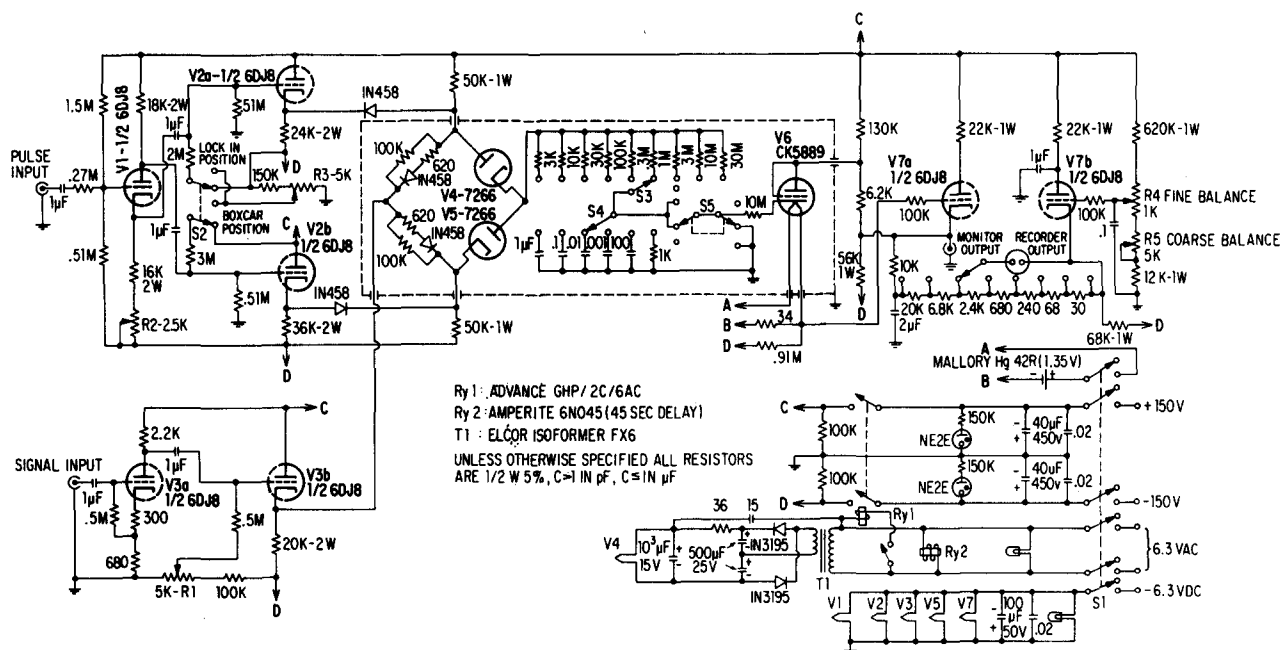


FIG. 8. Boxcar integrator circuit.

is amplitude dependent. For the moderate modulation indices used with this apparatus, the detector cutoff is roughly 0.5 Mc. The passband of the video stage can be varied from 250 cps to 0.6 Mc.

Overload recovery times (without a tuned receiver coil at the input) have been measured only for the preamplifier and amplifier connected together. With 10- μ sec bursts of rf up to 30 V at the preamplifier input, the recovery time is roughly independent of overload amplitude and is less than twice the value required by the fundamental limitation of the receiver bandwidth. When the video stage is set to its maximum bandwidth, the overload recovery times vary from 0.8–2.5 μ sec, depending on the bandwidth of the tuned rf stages.

I. Boxcar Integrator

The boxcar integrator (Fig. 8) is closely patterned after a design of Blume,⁸ but modified to fit the power supply voltages, pulse outputs, and other characteristics of our apparatus. Except for a maximum linear (to within 1%) output of 17 V and an over-all maximum gain of 1.0, the performance of this unit is essentially the same as that given in Ref. 8. The construction details of Ref. 8 have, except as noted below, been followed for our circuit.

Some features of Fig. 8 merit mention here. Relays 1 and 2 insure that all tube filaments have been brought to operating temperatures before the $B+$ and $B-$ supply voltages are applied. The positive pulse input of 60 V (supplied by the pulse generator chassis) is attenuated and converted to dual polarity at V1. Potentiometer R2 allows the pulses fed to the gate driver V2 to be balanced. For low duty factor operation, S2 is put in the boxcar position. Since the pulse rig is, in principle, capable of recording conventional crossed coil steady-state resonances, a lock-in position has been provided for S2. In this mode, a square wave is fed to the pulse input and R2 and R3 are used to symmetrize the driving waveform at the diode gate. The boxcar integrator is then capable of use as a conventional lock-in detector. Potentiometer R1 allows the cathode potential of V3b to be set at ground. Recorder output zero setting is accomplished with R4 and R5. Single turn carbon units have given satisfactory stability and resolution for R1, R3, and R4. Switches S3, S4, and S5, which must maintain high leakage resistance to ground, are ceramic rotary units.

J. Power Supplies

The pulse rig has power requirements roughly as follows: (a) -6.3 V dc at 15A, (b) 6.3 V ac at 10 A, (c) -150 V dc at 150 mA, (d) +150 V dc at 450 mA, (e) +500 to +800 V dc at less than 20 mA, and (f) +500 to +3,000 V dc at less than 15 mA. Voltage (a) is provided by a 6-V automobile

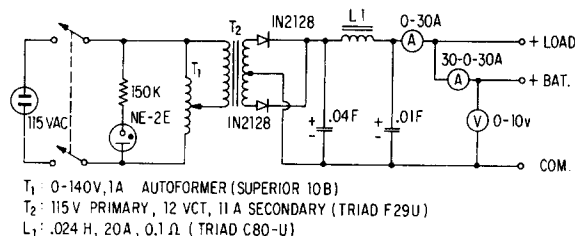


FIG. 9. Minus 6.3-V dc supply circuit.

battery continuously charged by the low ripple power supply diagrammed in Fig. 9. The auto-transformer in this supply is adjusted so that the net current drain on the battery is zero. A filament transformer driven from an auto-transformer is adjusted to give 6.3 V ac at the transmitter power tube sockets. It is important to keep the resistance in the leads connecting supplies (a) and (b) and the respective chassis low, otherwise the tube filament voltages will fall below acceptable values. Voltage (c) is provided by a Kron-Hite model UHR-225 power supply and (d) by a Kron-Hite model UHR-245. The transmitter screen supply (e) is a Power Designs, Inc., model 701M. Plate voltage for the output stage of the transmitter (f) is supplied by a John Fluke model 408A. A meter to monitor the power tube plate current has been placed between the power supply and the transmitter. The relatively large output impedance of this supply is compensated with the 4- μ F plate voltage capacitor in the transmitter.

An interlock system (not shown) has been included to prevent the application of the positive plate and screen voltages to the apparatus in the absence of the negative bias voltages from (a) and (c).

K. Transmitter and Receiver Coils

The transmitter and receiver coils are an important part of any pulse apparatus. Since these coils are often tailored to a specific experiment, the discussion presented here will not describe a particular set of coils, but will outline some of the more important features of "coil strategy."

We choose a medium to high Q parallel RLC circuit resonant at the nuclear resonance frequency ν_0 to represent the transmitter coil. A reasonable geometry for this coil is a single layer solenoid with a radius two-thirds its length. If the transmitter pulse power output to the coil is P (watts), the volume of the coil V (cm^3), the Q of the circuit Q , the 10–90% rise and fall time of the coil T_R (μ sec), and the peak amplitude of the induced oscillating rf magnetic field $2H_1$ (gauss), then

$$H_1 \approx 3(PQ/\nu_0 V)^{1/2} \approx 3.7(PT_R/V)^{1/2}. \quad (1)$$

Other formulas relevant to transmitter coil design are

$$2\pi\nu_0(LC)^{1/2} = 1, \quad (2)$$

⁸ R. J. Blume, Rev. Sci. Instr. 32, 1016 (1961).

and

$$2Q/3\nu_0 \approx T_R \approx 4RC. \quad (3)$$

Ordinarily, one seeks as large an H_1 as possible. The size of the sample determines V ; so the parameters left to maximize are T_R and P . The pulse power P is maximized by matching R to the output impedance of the transmitter and the free induction decay lifetime sets an upper limit on T_R . A rough rule for T_R is that it be not greater than $T_2^*/50$.⁹ When these choices have been made, (2) and (3) determine R , L , and C .¹⁰ As long as it is tuned to ν_0 , the performance of a transmitter coil is usually a slowly varying function of R , L , and C ; therefore, moderate variations from the values prescribed above are often tolerated.

An application of Eqs. (1), (2), and (3) to our transmitter is given here to indicate typical circuit parameters. Assume a transmitter coil volume of 5 cm^3 , a resonance frequency of 10 Mc, and a T_R of $0.25 \mu\text{sec}$. The pulse power output is 5 kW and R is 250Ω . These values give an H_1 of 60 G, a coil inductance of $1 \mu\text{H}$, a tuning capacitance of 250 pF , and a circuit Q of about 4. The peak rf voltage applied to the transmitter coil will be about 1.6 kV.

The requirements of the receiver coil are somewhat different from those of the transmitter coil. A receiver coil is designed to present the free induction decay signal with the highest possible signal-to-noise ratio. To illustrate which receiver coil parameters are important, we shall calculate this ratio. A sensible model is given by a high Q parallel RLC circuit at a temperature T_C feeding a receiver system with an equivalent shot noise resistance R_{eq} at a temperature T_{eq} and an infinite input impedance. This model is an adequate representation of our pulse apparatus if R_{eq} is 220Ω and T_{eq} is 300°K .

Under these conditions the signal-to-noise ratio at the receiver output can be increased by narrowing the frequency passband. However, the lifetime of the free induction decay signal imposes a lower limit on the receiver bandwidth (B) if severe distortion is to be avoided. This limit can, in principle, be applied anywhere in the system, including the receiver coil itself. In practice, it is almost always applied at the video stage of the receiver, which will be assumed in what follows.

The rms noise voltage $[E(\nu)]$ in a small frequency range $d\nu$ appearing across the terminals of an unloaded impedance $Z = R(\nu) + iX(\nu)$ at a temperature T , is¹¹

$$E(\nu) = [4kTR(\nu)d\nu]^{1/2}, \quad (4)$$

where k is Boltzmann's constant. For our parallel RLC

circuit,

$$R(\nu) = \frac{4kTR}{1 + [2Q(\nu - \nu_0)/\nu_0]^2}, \quad (5)$$

and

$$R = 2\pi\nu_0QL. \quad (6)$$

Since we assume B is much less than the receiver coil bandwidth (ν_0/Q), the output thermal noise (E_N) is

$$E_N = G[4kTB(2\pi\nu_0QLT_C + R_{eq}T_{eq})]^{1/2}, \quad (7)$$

where G is the receiver system gain.

The free induction decay signal can be separated into parameters representing the state of the nuclear spin system and parameters representing the coil. Nuclear spin properties will be lumped into a parameter α . Then, the signal voltage (E_S) at the output of the receiver is

$$E_S = G\alpha NAQ\phi, \quad (8)$$

where N is the number of turns of the receiver coil, A is the cross sectional area of the coil, and ϕ is the filling factor. Now,

$$\frac{E_S}{E_N} = \frac{\alpha NAQ\phi}{[4kB(2\pi\nu_0QLT_C + R_{eq}T_{eq})]^{1/2}}. \quad (9)$$

To proceed further we make a reasonable assumption about the coil, namely, that it is a single layer solenoid with a radius approximately equal to its length. The inductance of the coil is then

$$L = DA^{1/2}N^2, \quad (10)$$

where D is a slowly varying function of the coil shape. Substitution of (10) into (9) gives

$$\frac{E_S}{E_N} = \frac{\alpha NAQ\phi}{[4kB(2\pi\nu_0QDA^{1/2}N^2T_C + R_{eq}T_{eq})]^{1/2}}. \quad (11)$$

From this, it is clear that one wants T_C as low as possible.

We now separate E_S/E_N into two limiting cases; $R_{eq}T_{eq} \ll RT_C$, and $R_{eq}T_{eq} \gg RT_C$. The first is typical of experiments carried out at all but low temperatures. Since experiments performed below 4°K usually generate conditions intermediate between the two extremes, one then picks coil parameters that are an intuitive compromise between those suggested by the extremal cases. Now

$$\frac{E_S}{E_N} = \alpha\phi \left[\frac{AQ}{8\pi k\nu_0 DBT_C} \right]^{1/2} \quad \text{if } R_{eq}T_{eq} \ll RT_C, \quad (12a)$$

and

$$\frac{E_S}{E_N} = \frac{\alpha NAQ\phi}{(4kBR_{eq}T_{eq})^{1/2}} \quad \text{if } R_{eq}T_{eq} \gg RT_C. \quad (12b)$$

In both cases, a good filling factor is desired, which implies that the coil fits closely on the sample and is shorter than the sample. Making A large then suggests a large sample.

⁹ Available components or special experimental considerations often limit T_R to less than this value.

¹⁰ A useful reference for the design of high Q coils is *Radiotron Designer's Handbook* (Radio Corporation of America, Harrison, New Jersey, 1956), pp. 463-474.

¹¹ Lawson and Uhlenbeck, *Threshold Signals* (McGraw-Hill Book Company, Inc., New York, 1950), p. 74.

A large Q is favored for both cases, and in the latter, a large number of turns.

The reader may be surprised that the number of turns N does not appear in (12a). This happens because the noise is determined entirely by the resonance impedance of the receiver coil circuit. For a given Q and coil geometry, the resonance impedance is proportional to L , and thus to N^2 . The signal is proportional to N ; therefore, terms with N cancel in E_S/E_N .

Since the coil parameters Q , N , A , ϕ , and D are complicated functions of each other, it is difficult to derive exact statements to replace the qualitative ones given above. For many cases, it has been found empirically that to maximize E_S/E_N , a sensible prescription is to start with the largest sample possible, wind as close to the sample as possible the highest Q coil that can be made and tuned with the receiver, and let the other geometrical factors take care of themselves. When operating at low temperatures, placing most of the receiver coil tuning capacitance next to the coil in the low temperature bath will help maximize Q and reduce T_c .

This treatment of transmitter and receiver coils has considered them as independent entities. In fact, they are not, and unless some precaution is taken, voltages appearing on the receiver coil during a transmitter pulse can damage the receiver or adversely affect its operation. The solution of this problem has led to the three kinds of coil arrangements commonly used with pulse setups.

Since these schemes have unique virtues and defects, and can be used with our apparatus, they will all be described here. (Many specific designs are given in the literature.) A geometrical solution is to mount the (separate) receiver coil inside the transmitter coil with their axes orthogonal to minimize inductive coupling. As only the component of H_1 perpendicular to the externally applied magnetic field (H_0) is effective in tipping the component of the nuclear magnetization initially parallel to H_0 (M_z), it is customary, but not necessary, to put the axis of the transmitter coil perpendicular to H_0 . Similarly, the free induction decay signal is maximized if the axis of the receiver coil is perpendicular to H_0 . For these reasons, one usually makes the axes of the two coils and H_0 mutually perpendicular. This crossed-coil arrangement is the one most commonly used with our apparatus.

An electrical solution (rf bridge) is to use the same (sample) coil for transmitter and receiver, connect the transmitter output to drive it plus an identical (dummy) coil without any sample, and connect the receiver to register the voltage difference between the sample and dummy coils. The sample coil is usually mounted perpendicular to H_0 . When this system is properly adjusted, the rf pulse is nulled at the output, but the free induction decay appears as a difference signal between the two coils. To employ this system with our apparatus requires the

addition of a push-pull rf transformer at the transmitter output or at the receiver input.

The third method (weak coupling to nonlinear input) is to couple a single transmitter-receiver coil through a moderate impedance to a high input impedance receiver which has been protected by a limiting element, such as parallel crossed silicon diodes.¹² Again, the axis of the coil is usually made perpendicular to H_0 . With this scheme, the limiting action at the receiver input causes the large transmitter voltage to be dropped across the coupling impedance, but most of the subsequent free induction decay signal appears at the receiver input, for it is insufficient to initiate the limiting action there.

Even though these coil arrangements will protect the receiver from damage by the rf pulse, they often permit the transmitter pulse to cause troublesome interference with the subsequent free induction decay signal. This interference can come from inductive coupling to the transmitter coil during the latter stages of its electrical ringing decay or from electrical ringing in the receiver coil which has been induced by the shock of the transmitter pulse. These electrical ringing signals must decay to values much less than the free induction decay amplitude (often on the order of microvolts) before they cease to interfere. The ringing times will, of course, be reduced if the Q 's of the respective coils are reduced. Such a reduction may conflict with the requirement of a large H_1 in the transmitter pulse and/or a high E_S/E_N for the receiver coil. In the case of our apparatus, the problem with the transmitter coil has been alleviated by making its Q large for large signals and small for small signals. This is accomplished with the series connected 6V3A diodes in the transmitter. It speeds the recovery of the transmitter coil without reducing appreciably the available H_1 during a pulse.

Just the opposite has been done for the receiver coil with the quench circuitry. Here, the Q of the receiver coil is lowered during and for a short time after the pulse to speed the disappearance of the ringing, then raised to record the free induction decay signal with a more favorable signal-to-noise ratio. It is easily seen that the time sequence of these operations makes them inapplicable for a combination transmitter-receiver coil unless additional gating operations are added to the apparatus.¹³ Therefore, we use this feature only with crossed coils.

Since one decision about coil design is what transmitter-receiver coil configuration to use, the merits and demerits of the different arrangements will be briefly mentioned

¹² This weak coupling can be imposed by a resistance, inductance, mutual inductance, capacitance, nonlinear element with a high impedance for large signals and a low impedance for small signals, or combinations thereof.

¹³ A Q gating circuit that could withstand the large transmitter voltage and be operative for short times after a pulse would be useful with combination transmitter-receiver coils. An example applicable to our apparatus would be to gate the -150 -V bias on the 6V3A tubes in the transmitter.

here. The advantages of the crossed-coil configuration are that (a) conflicting transmitter and receiver coil requirements are easily met, and (b) a single-ended output from the transmitter is sufficient. On the other hand, it is (a) not possible to do pure quadrupole resonance experiments with orthogonal crossed coils,¹⁴ (b) placement of the transmitter outside the receiver coil is wasteful of H_1 , and (c) placing two coils around the sample complicates experiments performed under special conditions, such as high temperatures, low temperatures, high pressures, in vacuum, or in special atmospheres, etc. Fortunately, with our apparatus, careful crossing of the coils is not necessary. It is sufficient merely to cross them by visual inspection. The performance of the pulse apparatus is not hampered by signals as high as 30 V on the receiver input during an rf pulse.

Advantages of the rf bridge configuration are (a) experiments performed in special environments are relatively easy, as only one coil surrounds the sample and (b) pure quadrupole resonance experiments are possible. Some disadvantages of this system are (a) the bridge balancing adjustments are often difficult and subject to drift, (b) a push-pull transmitter output or receiver input is required, (c) H_1 in the dummy coil is wasted, (d) the dummy coil reduces the signal-to-noise ratio by loading the sample coil, and (e) the single sample coil cannot be simultaneously optimized for possibly conflicting transmitter and receiver coil requirements.

The advantages of a single coil with weak coupling to a nonlinear input are (a) the simplification of a single coil, (b) the ability to do pure quadrupole resonance, (c) the need of only a single-ended transmitter output, and (d) no wasted H_1 . These virtues are balanced by (a) an inability to simultaneously optimize the functions of a receiver and transmitter coil and (b) a possible loss in signal-to-noise ratio forced by the presence of the weak coupling network.

With all of these coil arrangements, it is good practice to shield the coils; otherwise, they easily pick up interference broadcast by external sources. It has not been necessary to shield the coils to prevent our receiver from breaking into oscillation.

Finally, we point out a coil "disease" commonly found in pulsed magnetic resonance experiments. The symptoms are a high Q spurious ringing signal which follows an rf pulse and whose amplitude increases with H_1 and H_0 . This ringing can persist for milliseconds and completely overwhelm a small free induction decay signal. We believe it is caused by exciting acoustical modes near the master oscillator frequency within the coils and/or sample.

It is not surprising that such oscillations should exist. In the case of the transmitter coil, one has circulating rf currents on the order of 50 A which, in the presence of a

large magnetic field, impose large oscillatory driving forces on the transmitter coil. Since the wavelength of acoustical vibrations in the coil materials is typically on the order of the diameter of the coil wire at commonly used frequencies, it is reasonable that acoustical modes of the coil material should overlap the driving frequency. Given these oscillations, it is easy to postulate mechanisms for coupling them to the receiver coil.

These unwanted signals can be reduced or eliminated by either not exciting them and/or lowering their Q . One way, which works part of the time, is to employ a different nuclear resonance frequency. Another occasionally successful treatment is to find an rf pulse length which minimizes the ringing signal. Careful crossing of transmitter and receiver coils will sometimes eliminate this signal. The ringing can often be avoided by changing the wire diameter of the transmitter coil, using finely stranded wire for the transmitter coil, using tinned wire for the transmitter coil, coating the transmitter coil with an acoustically lossy substance, or immersing the transmitter coil in an acoustically lossy dielectric liquid. These same prescriptions apply to spurious ringing signals emanating from the sample. No single procedure has been found uniformly successful at eliminating this ringing; however, it is usually possible to find some combination of the above treatments that will make it negligible.

III. OPERATION

A. Introduction

The nuclear resonance parameters which determine the optimum settings of the pulse apparatus can vary so widely that no single prescription for setting up the apparatus can be given. In Sec. III B, we shall, therefore, describe the optimum setup procedure for observing a "typical" free induction decay; i.e., one for which the free induction decay lifetime (T_2^*) is 10^{-3} sec, the spin-lattice relaxation time (T_1) is 10^{-1} sec, the length of a 90° pulse (τ) is $10 \mu\text{sec}$, the resonance absorption lineshape is symmetrical, the frequency width of the resonance is small compared to the resonance frequency, and the signal-to-noise ratio of a free induction decay is greater than three. If the properties of the resonance to be observed are not known, our prescription is still useful for an initial search. The reader's understanding of nuclear resonances will be relied upon for making final pulse adjustments once the resonance is under observation and for coping with situations not described here. We also assume that a suitable crossed-coil system is in use.

Section III C describes how to generate varieties of pulse trains, and Sec. III D covers the use of the boxcar integrator, including the method for recording $\chi''(\Omega)$ and $\chi'(\Omega)$ directly. In both of these sections, the restrictions on nuclear resonance parameters prescribed for Sec. III B

¹⁴ T. P. Das and E. L. Hahn *Nuclear Quadrupole Resonance Spectroscopy Supplement 1 to Solid State Physics*, edited by F. Seitz and D. Turnbull (Academic Press Inc., New York, 1958) p. 74.

apply except for a relaxation of the conditions on τ , T_2^* , and T_1 to $\tau \ll T_2^* \ll T_1$.

B. Typical Setup Procedure

First, make all the connections shown in Fig. 1 and switch the transmitter and receiver to the correct frequency band.

For the initial warmup, turn on the supplies providing -6.3 V dc, 6.3 V ac, -150 V dc, and $+150$ V dc in that order. The -6.3 -V dc supply should be adjusted to give no net current through the storage battery. The 850-V screen and high voltages will be turned on later. When turning off the apparatus, invert the order for turning it on. Set the master oscillator to the desired frequency and adjust its output to 6.0 V pp (volts peak-to-peak). The square-wave generator frequency should be set to 10^3 cps and its amplitude to 9.0 V pp.

On the phase shifter-attenuator chassis the phase shift delay lines should be set for a minimum delay. The fine reference amplitude control is set to its maximum reference output position, and the coarse reference amplitude control is set to its minimum reference output position.

The pulse generator controls are set as follows: Channel 1 (S2) to NC and channel 2 (S6) to NC. Channel 3 (S9) is set to the target zero position with a delay of 100 μ sec and a width of 10 μ sec. (The purpose of the delay is to establish a baseline on the oscilloscope before the arrival of the rf pulse.) Switch channel 4 (S12) to the target 9 position and introduce a 0.1 -sec delay between pulse trains. Now close (momentarily) the beam form switch (S15). This operation should register an increase of a few mA in the current from the $+150$ -V supply. Set the oscilloscope to trigger externally on channel 3 (S1), adjust its sweep speed to 100 μ sec/cm, and its vertical gain to show the noise output from the receiver. The 150 -k Ω rheostat in the quench control circuit is set to its minimum value.

Now turn on the screen voltage (850 V) and the high voltage, which should initially be set to 2 kV. Set switch S1 to give the highest screen voltage. The transmitter can be tuned by observing its output on an oscilloscope with a $100:1$ attenuator probe or by monitoring the current drawn from the high voltage supply. With oscilloscope monitoring, first the ganged capacitors of the driver stages and then the transmitter coil tuning capacitors are adjusted to maximize the rf voltage on the transmitter coil. When using the plate current as a monitor for tuning, first set the duty factor to about 0.005 . This can be done, for example, by temporarily setting channel 4 to a delay of 0.01 sec and the rf pulse width to 100 μ sec. The driver stages are tuned by maximizing the current from the high voltage supply with respect to settings of the ganged tuning capacitor, and the transmitter coil is tuned by minimizing this current with respect to settings of the

transmitter coil tuning capacitors. It is important that during this tuneup procedure, or at any time, the duty factor not be allowed to go so high that the steady-state plate current to the output stage exceeds about 10 mA. At the higher frequencies, it may be necessary to trim the tank coils for maximum power output.

After the transmitter has been tuned, the pulses should be set back to their previous configuration. If maximum power output is desired, the high voltage can be advanced to 3.0 kV.

The next step is to adjust the receiver. Set the gain control to AGC, the positive peak clipper to the maximum output position, and the bandwidth switch to the maximum bandwidth position. Tune the receiver for maximum gain by monitoring the rf output with an oscilloscope or a vacuum tube voltmeter. If, during this step, there is evidence of interference from AGC action, reduce the amplitude of the reference signal. After tuning the receiver, return the reference signal to its former amplitude. Set the 170 - Ω attenuator for no attenuation.

Now the receiver coil must be tuned. This can be done approximately by adjusting the tuning capacitors in the preamplifier (C1,C2) to give a maximum noise output from the video output of the receiver (V5b). A more precise method of tuning the receiver coil is to remove the reference signal from the receiver and connect it temporarily to the tune input of the preamplifier. Then, while using the 170 - Ω attenuator to keep the reference signal below the AGC threshold, C1 and C2 are adjusted to maximize the amplitude of the reference signal at the rf output of the receiver. After this is done, the reference signal should be returned to the receiver and the 170 - Ω attenuator reset to its previous value. The receiver coil tuning should be checked occasionally during the course of long runs.

The next step is the adjustment of the quench control waveform duration. Figure 10 illustrates the effect of this adjustment. In each picture, the upper trace depicts a typical rf pulse overload waveform at the receiver coil, and the lower trace is the simultaneous rf output from the receiver. The rf pulse causes an overload signal of 4 V followed by a ringing decay at the input of the preamplifier. At the receiver output, limiting of these signals is observed until the input has dropped to about 1 mV. A small unmodulated rf signal has been inductively coupled into the receiver coil to demonstrate what would be the effect of the variation of receiver coil Q on a free induction decay. Since the frequency of this unmodulated signal is slightly different (about 50 kc) from the unmodulated reference signal, addition of the two signals in the receiver produces the beats appearing in the rf output. In Fig. 10(a), where the duration of the quench control waveform is too short, the receiver coil rings for a time determined by its unloaded Q . The ringing persists long enough to overload the

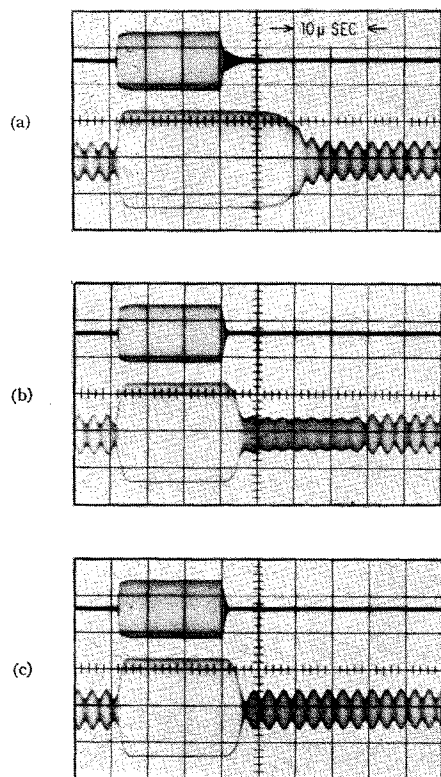


FIG. 10. Receiver coil recovery waveforms. In each picture, the upper trace shows an rf pulse overload waveform at the input to the preamplifier with a sensitivity of 5 V/cm, and the lower trace is the rf output from the receiver with a sensitivity of 2 V/cm. The sweep speed is 5 $\mu\text{sec}/\text{cm}$ and the graticule is 6×10 cm. Beats with the 10-Mc reference signal are provided by a 150- μV unmodulated signal inductively coupled to the receiver coil. These beats demonstrate the effects of overloading on a nuclear resonance signal. In (a), no quench signal is fed to the preamplifier and receiver coil ringing overloads the receiver for about 14 μsec . At (b), where the duration of the quench signal is too long, the sensitivity of the receiver coil is impaired for 20 μsec . In (c), a properly adjusted quench signal provides an optimum recovery from ringing of about 5 μsec .

receiver for about 14 μsec after the end of the rf pulse. In Fig. 10(b), the duration of the quench waveform has been made too long. Even though the receiver overload period is short, the lowered Q of the receiver coil severely reduces the signal for about 20 μsec after the end of the rf pulse. The correct adjustment [Fig. 10(c)] is that point at which increasing the duration of the quench control waveform just ceases to shorten the overload limiting time in the receiver. Here, a recovery time of 5 μsec after the end of the rf pulse is demonstrated.

The apparatus is now ready for the preliminary observation of our "typical" free induction decay signal, an approximate example of which appears in Fig. 11. Each of the upper traces is from the video output of the receiver and each of the lower traces is from the rf output of the receiver. In Fig. 11(a), an off-resonance condition is shown; i.e., the magnetic field has been adjusted so that the nuclear resonance frequency is slightly different from

the reference frequency, producing beats in the output. The corresponding on-resonance condition is shown in Fig. 11(b), where the phase shifter has been adjusted to give a positive maximum for the initial part of the video output waveform, and the positive peak clipper has been adjusted to make its action visible.

At this point, the on-resonance free induction decay signal should be maximized with respect to adjustments of the phase shift control and receiver coil tuning. The adjustment of the quench control should be checked. If necessary, the positive peak clipper should be set to prevent saturation of the oscilloscope amplifier. The reference amplitude should be set so that the free induction decay modulates the reference signal by no more than 30% of the latter. When the free induction decay signal is huge, this criterion may require use of the 170- Ω attenuator. Ordinarily, the receiver coil damping resistors are not used. If, however, a large bandwidth or extra fast overload recovery are needed, it may be necessary to switch them in (at the loss of signal-to-noise ratio).

Further adjustments to be made are in the rf pulses.

C. Pulse Configurations

It is customary to set the rf pulse lengths to give 180 or 90° rotations of the nuclear magnetization by observing the effect on a free induction decay signal. A 180° pulse, which inverts the nuclear magnetization, is obvious by the lack of a subsequent free induction decay.

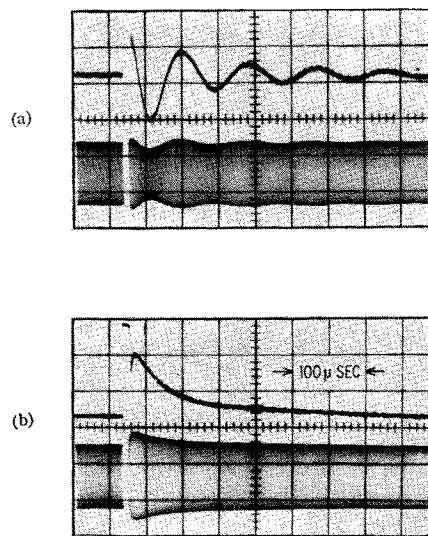


FIG. 11. Free induction decay waveforms. The ^{127}I free induction decay signal from a single crystal of KI is shown. In each picture, the upper trace records the output of the video stage and the lower trace the rf output of the receiver. Photograph (a) shows an off-resonance signal, and (b) the on-resonance signal. The phase shift control has been adjusted to give a maximum on-resonance signal, the sweep speed is 50 $\mu\text{sec}/\text{cm}$, and the graticule size is 6×10 cm. The 12- μsec interruption in the waveforms is the duration of the rf pulse overloading and recovery. The limiting level of the positive peak clipper has been lowered to make its action visible in the top trace of (b) during the rf pulse overload period.

If one starts from thermal equilibrium, a single 90° pulse will maximize the on-resonance free induction decay signal. This operation is usually done with channel 3. An alternative method, which can often be used with channel 2, is to produce two (targets 8, 9) equal width pulses closely spaced by a high square-wave generator frequency. Adjustment of the pulse width for a null free induction decay signal after the second pulse gives 180° for the pair, or 90° for each. If target 8 is then switched out, a single 90° pulse remains. This 90° pulse can be transferred to another channel by substituting the other channel for the target 8 position and again adjusting for a null or by simply setting up another pulse with the same length. The ambiguity between a pulse of angle θ and a pulse of angle $\theta + 360^\circ$ can be removed by observing the additional pulse length required to rotate M_z by an additional 90 or 180° .

Pulses of other angles are usually set up by scaling their length from the known length of a 90 or 180° pulse. A pulse of about 8° , which leads to a free induction decay amplitude $1/7$ that of a 90° pulse but reduces M_z by only 1% , is usually best generated by reducing H_1 . This is accomplished by reducing the transmitter screen voltage (S1) and/or loading the transmitter coil. To produce such small angle pulses by reducing their length may lead to unreasonably short pulses.

Most pulsed-resonance measurements involve combinations of pulses whose length can be set according to the above prescriptions. The rest of this section will describe ways to set up several multiple pulse combinations on our apparatus.

One of the most commonly used sequences is a pair of pulses, often $180-90^\circ$ or $90-90^\circ$, whose spacing is varied to measure T_1 . Let us assume the $180-90^\circ$ pair is to be used. Channel 4 is triggered by target 9 and is set for a delay of $5T_1$ or greater. The 90° pulse is generated in channel 3, which is initially triggered by target 9 and adjusted for a delay of about T_2^* . Channel 3 also triggers the oscilloscope, whose horizontal sweep should last for $5T_2^*$. The 180° pulse is generated in channel 2. Here, the delay is adjusted so that the 180° pulse just ends as the 90° pulse begins. The period of the square-wave generator should be set at about $\frac{1}{4}T_1$. Then, switching channel 3 to trigger on different targets will display the amplitude of the free induction decay as M_z recovers from inversion. The recovery times are integral multiples of the square-wave generator period. Any other set of pulse pairs starting from thermal equilibrium whose spacing is to be varied can be handled in the same way. Continuous variation of the spacing between the pair can be obtained by sweeping the frequency of the square-wave generator or sweeping the duration of one of the delay generators.

The 90° -multiple 180° pulse sequence of Carr and Purcell,² which can be used to measure T_2 for the case of spacial diffusion in a magnetic field gradient, is produced

as follows: The oscilloscope sweep duration is adjusted to $3T_2$ and is triggered from channel 3. Channel 4 is triggered from target 9 and its delay is set for $5T_1$ or greater. The 180° pulses are generated by channel 2, which is triggered from targets 1-9 and set for a minimum delay. A square-wave generator period of roughly $\frac{1}{3}T_2$ is introduced. Channel 4, triggered by target O, generates the 90° pulse. Its delay is then adjusted so that the spacing between the 90° pulse and the first 180° pulse is one-half the spacing between 180° pulses.

The generation of many other pulse sequences, such as a picket of 90° pulses¹⁵ or multiple saturation pulses followed by a 90° pulse,¹⁶ is possible with simple modifications of the prescriptions given above.

D. Boxcar Integrator

For this discussion the following simple model will be chosen to represent the action of the boxcar integrator. The output of the receiver is applied across a series combination of an ideal switch (boxcar gate), a resistor (S3, Fig. 8) of resistance R , and a capacitor (S4, Fig. 8) of capacitance C . The switch is closed during the boxcar gate pulse of width w and open at all other times. The output of the boxcar is taken across C . This model is a good representation of the circuit of Fig. 8 for almost all ways in which it can be used.

Since the output of our model boxcar will change only during a gate pulse, its evolution in time is most conveniently represented by a reduced time variable t , which progresses normally during a boxcar pulse and is stationary between pulses. Therefore, the response of the recorder is well approximated by the potential held on C between boxcar pulses.

We shall assume that the boxcar is being used in the integral mode, that is, the product RC is made large enough compared to w so that the current charging C is proportional to the average value of the receiver output during the boxcar pulse. The error in this assumption will not exceed about 2% if $RC \gtrsim 4w$. It is, of course, possible to use the boxcar in other ways, but they shall not be described here.

The pulses to operate the boxcar are taken from channel 1 of the pulse generator. (Switch S5 of Fig. 4 should be open.) It is assumed that all voltage offsets have been adjusted so that, in the absence of an input to the boxcar, the voltage across C is zero with the boxcar pulses operating.

The conventional way to measure T_1 with a boxcar is to use a $180-90^\circ$ pulse sequence and sample a small portion of the video output with an on-resonance free induction de-

¹⁵ E. G. Wikner, W. E. Blumberg, and E. L. Hahn, Phys. Rev. 118, 631 (1960).

¹⁶ W. Gilbert Clark, Bull. Am. Phys. Soc. 7, 481 (1962).

cay. This is done for a fixed number of times as a function of the spacing between the 180 and 90° pulses. A recorder deflection proportional to the amplitude of the video output signal is then registered. If a small number of boxcar pulses is used to sample for each 180-90° pulse separation, the system responds to the initial charging rate of C , and C should be discharged when changing the 180-90° pulse separation. If, however, C is allowed to charge for a number of pulses greater than about $4RC/w$, a steady-state output will be reached and one can change the 180-90° pulse spacing without discharging C .

The 180-90° pulses are generated as described in Sec. III C. Channel 1, which supplies the boxcar pulse, is set to trigger on channel 3 (position X of S2, Fig. 4). The delay and width controls of channel 1 are used to pick out the desired segment of the video signal. The timing of this operation can be observed on the oscilloscope if the video signal and boxcar driving pulse are simultaneously displayed. Since, in this configuration, the channel 1 delay is triggered at the beginning of the 90° pulse, the boxcar pulse accurately tracks all changes in position of the 90° pulse. Changes in the spacing between the 180 and 90° pulses can then be accomplished in the usual way.

Another standard boxcar operation is to record the shape of a free induction decay. This is done by sweeping a narrow sampling pulse through the video signal from an on-resonance free induction decay generated from thermal equilibrium by a 90° pulse. The setup procedure is the same as for 180-90° T_1 measurements except that the 180° pulse is eliminated. Channel 1 is directly triggered from the same target as channel 3, and the channel 1 delay is varied to sweep the boxcar pulse through the free induction decay. This sweep can be supplied by an externally driven potentiometer, as described in Sec. II E. It is important that the sweep not proceed so quickly that the boxcar is unable to follow the shape of the video signal. A rough criterion for avoiding this difficulty is to require that

$$n \gtrsim 50RC(fw)^{-1}, \quad (13)$$

where f is the error of the recorder trace in percent and n is the number of boxcar pulses required for any portion of the video signal that changes by $\frac{1}{2}$ the maximum value of the video waveform.

Since the boxcar output averages over the duration w , a faithful reproduction of the video output waveform will result only if w is much less than the shortest time interval over which the waveform changes appreciably.

This type of boxcar sweep will also record the shape of spin echoes.

The boxcar procedures described so far use only a small portion of each free induction decay and discard the rest, a procedure that wastes much available information. We shall now briefly describe what happens if the boxcar pulse

covers the entire free induction decay signal.¹⁷ The method to be described requires phase coherent detection.

Let the free induction decay signal be $S(t)\cos(\omega_0 t)$, where time (t) is measured from the end of the 90° pulse. The resonance frequency is ω_0 and $S(t)$ is the envelope of the decay waveform. After this signal has been added to the larger amplitude reference signal of frequency ω and demodulated by an ideal peak reading detector, the video output signal is proportional to $S(t)\cos[(\omega - \omega_0)t - \varphi]$, which we rewrite as $S(t)[\cos(\Omega t)\cos\varphi + \sin(\Omega t)\sin\varphi]$. The phase difference φ is controlled by the rf phase shifter.

When the boxcar pulse width w covers the entire free induction decay and $RC \gg w$, the steady-state response $V(\Omega)$ of the boxcar is

$$V(\Omega) = K \int_0^\infty S(t)[\cos(\Omega t)\cos\varphi + \sin(\Omega t)\sin\varphi] dt, \quad (14)$$

where K is a factor to take care of the gain of the apparatus. The range of the integral can be extended to ∞ because of the finite lifetime of the free induction decay. For many (and perhaps all) free induction decay signals,¹⁸ it can be shown that the steady-state unsaturated absorption $\chi''(\Omega)$ is proportional to the Fourier cosine transform of the free induction decay envelope and the corresponding dispersion $\chi'(\Omega)$ is proportional to the Fourier sine transform of the free induction decay envelope. In such cases,

$$V(\Omega) = K'[\cos\varphi\chi''(\Omega) + \sin\varphi\chi'(\Omega)], \quad (15)$$

and K' takes care of the constants. In general, a mixture of $\chi''(\Omega)$ and $\chi'(\Omega)$ is obtained. Setting $\varphi = 0^\circ$ or $\varphi = 90^\circ$ allows one to pick out either $\chi''(\Omega)$ or $\chi'(\Omega)$. A slow sweep of the magnetic field through resonance then allows one to record directly $\chi''(\Omega)$ or $\chi'(\Omega)$ at the output of the boxcar.

With this method, it is possible to use the pulse apparatus in a search for resonances by sweeping the magnetic field. If the parameters of the resonance are unknown, one has to guess at the correct pulse and boxcar settings. Even if these settings are grossly erroneous, some indication of the resonance will usually appear. Once the resonance has been found, the apparatus can be readjusted to provide a faithful reproduction of $\chi''(\Omega)$ and $\chi'(\Omega)$. The boxcar pulse length w generated in channel 1 is adjusted to cover the entire free induction decay, and the delay control is set so that the boxcar pulse begins as soon as recovery of the receiver is complete. An integration time constant $RC \gtrsim 4w$ insures that the integral of the video signal is taken.

The next step is to set φ . Several methods are available. If the video signal on the oscilloscope is not obscured by

¹⁷ To the best of the author's knowledge this scheme has not been reported elsewhere. For this reason, a more complete treatment of the method covering such points as ultimate sensitivity, distortions, and applications is planned for future publication.

¹⁸ I. J. Lowe and R. E. Norberg, *Phys. Rev.* **107**, 46 (1957); A. Abragam, *The Principles of Nuclear Magnetism* (Oxford University Press, London, 1961), pp. 98-103.

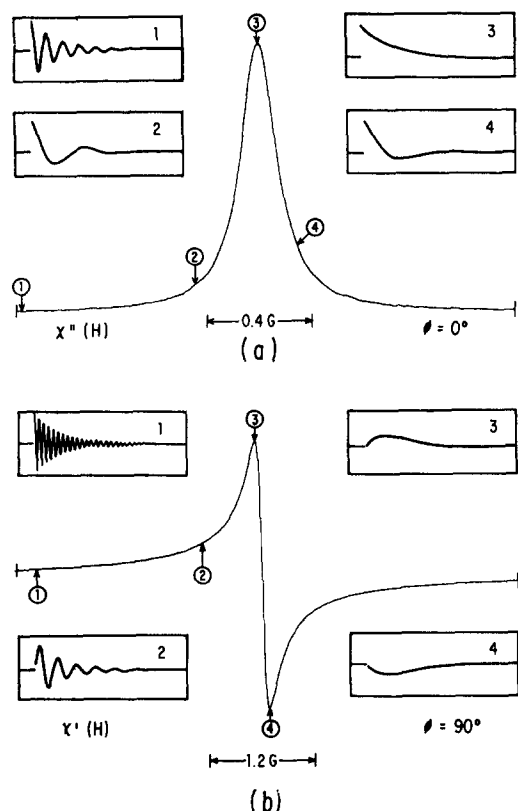


FIG. 12. Recorder-boxcar traces of $\chi''(\Omega)$ and $\chi'(\Omega)$ for the proton resonance of water doped with FeCl_3 . In (a) is shown a recorder trace of the boxcar output as a function of magnetic field for no initial phase difference (φ) between the reference signal and the free induction decay. In (b), the recording is shown for $\varphi = 90^\circ$. Phase-referenced free induction decay shapes (obtained from tracings of oscilloscope photographs) corresponding to points on the recorder traces are also given. Further experimental conditions are given in the text.

noise, φ can be set by noting that the on-resonance video signal is a maximum for $\varphi = 0^\circ$ and a null for $\varphi = 90^\circ$. An alternative method is to calibrate the rf phase shifter and then set it to whatever value is desired. If no such calibration has been made and the resonance is too weak for observation on the oscilloscope, φ can be set by referring to

the symmetry of the recorder trace; the functions $\chi''(\Omega)$ and $\chi'(\Omega)$ are, respectively, even and odd in Ω .

It is important to sweep through the resonance slowly enough that the boxcar has time to respond to the video signals. This constraint is expressed approximately by Inequality (13) if n is now interpreted as the number of boxcar pulses required for any portion of $\chi''(\Omega)$ or $\chi'(\Omega)$ that changes by $\frac{1}{2}$ its maximum value.

Figure 12 shows a resonance observed by this method. The sample used was protons in water doped with FeCl_3 , for which $S(t)$ should be an exponential function and $\chi''(\Omega)$ a Lorentzian.¹⁹ Video signals corresponding to selected points on the recorder traces are included. A master oscillator frequency of 10.0 Mc was employed. Ninety degree pulses of $\sim 2 \mu\text{sec}$, a boxcar sampling pulse width of 2.2 msec, a boxcar RC of 10 msec, a pulse repetition rate of 13 sec^{-1} , and a magnetic field sweep of 10 mG/sec for (a) and 30 mG/sec for (b) were used. The known proton gyromagnetic ratio was used to convert from the magnetic field to Ω . To an accuracy of less than 1% of its maximum value, the on-resonance free induction decay shape is exponential. Similarly, the shapes of $\chi''(H)$ and $\chi'(H)$ correspond to a Lorentzian $\chi''(H)$ to better than 1% of their maxima. Values of T_2^* obtained from the free induction decay envelope and $\chi''(H)$ agree to within 2%. These results verify Eqs. (14) and (15) for $\varphi = 0$ and $\varphi = 90^\circ$.

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¹⁹ A. Abragam, *The Principles of Nuclear Magnetism* (Oxford University Press, London, 1961), pp. 296, 328–330, 441.